EE426 / 626 Microprocessor Applications

Microprocessor / Microcontroller Project Development



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- Anthony E. Turvey
- LOA Department of Electrical and Biomedical Engineering
- Office: SEM 254E
- Office Hours: Monday and Friday 1:00 – 3:00pm
- Phone: N/A
- Email: aturvey@unr.edu

EE426 / 626 Microprocessor Applications

Lectures / Lab

- SEM 261 → SEM 341, Wednesday, 5:30 pm – 8:15 pm (capstone type project work)
- Lectures may be recorded and/or photographed with university permission
- WebCampus https://unr.edu/webcampus

- Syllabus and announcements will be posted on WebCampus
- It is your responsibility to ensure you have notifications turned ON (please)

- Capstone lab will be open and available to you at any time
- Scheduled courses take priority

EE426 / 626 Microprocessor Applications

Prerequisites

CPE301 – Embedded Systems Design
C/C++ Programming
Some proficiency with electronics and prototype development



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This is a capstone type project development course

CPE 301 Embedded Systems Design and C/C++ programming is assumed

Teams of 3 – 4 members each:

1. Project Manager
2. Technical Lead / Hardware Developer
3. Hardware / Firmware Developer
4. Firmware / App Software Developer

Red Teams (1 – 5) and Blue Teams (1 – 5) with Optional Team / Company Name

Project Presentations and Peer Evaluations



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Required Textbook

The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors, 3rd ed., Joseph Yiu, Newnes (an imprint of Elsevier Inc.) Waltham, MA, 2014

Required Materials / Resources



Linux, Windows, or Mac Computer or Laptop

Nucleo Development Board (STM32F446, Provided)

ST-Link Development Probe (Provided)

Optional: Segger J-Link Development Probe (Provided)

EE426 / 626 Microprocessor Applications

Use of AI In This Course is Allowed and Encouraged

- You may use AI to learn, check your understanding, but Not to Do Your Work
- Use AI tools as you would use: personal trainer, textbook, tutor, instructor office hours

- Be aware that AI tools can (and will) “hallucinate” and can potentially mislead you
- You are ultimately responsible for discerning the veracity of any AI generated content
- You are Not to Submit AI Output As Your Own Work – You must cite if and when AI is used

Guiding Principle – Ask Yourself the Following Question

Could I explain and reproduce this work on a blank sheet of paper, without the assistance of AI?

If the answer is NO, then the use of AI is doing you a disservice and is therefore not appropriate.

EE426 / 626 Microprocessor Applications

Points Distribution

Project Proposal:	10%
Prelim Design Review:	20%
Critical Design Review:	20%
Project Presentation:	20%
Peer Review:	10%
Individual Role:	20%
Total:	100%

Grading Scale

A:	93% - 100%	C:	74% - 76.9%
A-:	90% - 92.9%	C-:	70% - 73.9%
B+:	87% - 89.9%	D+:	67% - 69.9%
B:	84% - 86.9%	D:	64% - 66.9%
B-:	80% - 83.9%	D-:	60% - 63.9%
C+:	77% - 79.9%	F:	<60%

EE426 / 626 Microprocessor Applications

Special Accommodation:

Students may request accommodation for reason of military service, religious belief, disability, or any other special circumstance. Please make an appointment with your instructor to discuss any accommodation you may require to succeed in this course.

Academic Integrity

Departmental policy will be strictly enforced

Collaboration is required – Of course cheating is not!

Classroom Etiquette:

Arrive to class on time

Please silence cell phone

No distracting conversations

If you are sick, please be considerate of your fellow students



EE426 / 626 Microprocessor Applications

Week 2 PROPOSAL **10%**

Team formation, project description, peripheral checklist, BOM, Gantt chart, Risk assessment with fallback scope

Project disclosure (approval required to proceed)

Week 6 PDR (Preliminary Design Review) **20%**

Whole-class review, ~30 min/team, review-material handout before or at PDR

Week 11 CDR (Critical Design Review) **20%**

Whole-class review, ~30 min/team, closure of any open PDR action items + review-material handout before or at CDR

Week 14 FINAL PROJECT + PRESENTATION **20%**

Capstone presentation format, with Collateral:

Marketing brochure, Technical Specification, User's manual, Quick-start guide (mapped to team roles)

┌ PEER EVALUATION **10%**

Week 14 Structured peer-evaluation form

└ INDIVIDUAL ROLE DELIVERABLES **20%**

Graded per-member versus declared role

EE426 / 626 Microprocessor Applications

Project Manager

- Project Proposal
- Marketing Brochure
- Gantt Chart (Schedule)
- Risk Assessment and Fallback Plan
- Reporting Project Status, Schedule, Budget, and Meeting Minutes / Action Item Tracking

Technical Lead / Lead Hardware Developer

- User's Manual / Quick Start Guide
- Technical Specification
- Hardware Design and Development, Version Control, and Issues Tracking
- Preliminary and Critical Hardware Reviews and Follow-up

Software Developer (1 – 2)

- Code Design and Development, Version Control, and Issues Tracking
- Preliminary and Critical Software Reviews and Follow-up

